

GWELDJGBWAJHKDGF AWJKCD AWS

Alan Jung

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A decorative horizontal separator composed of twenty-five identical stylized flower icons arranged in a single row.

## 1 Cooper Pairs

Remember that  $\epsilon$  is the energy with respect to the Fermi energy. Last time we discussed how two electrons in a system would not automatically form a bound state. However, if we come sufficiently close to the Fermi level, they could! Such a **bound state** can be defined as the following superposition:

$$|\mathbf{k}, \uparrow; -\mathbf{k}, \downarrow\rangle \equiv |\mathbf{k}\rangle. \quad (1.1)$$

Denoting our free electron Hamiltonian as  $\mathcal{H}_0$ , we can say that it holds the following relation with our newly defined bound states:

$$\mathcal{H}_0 |\mathbf{k}\rangle = 2\epsilon(\mathbf{k}) |\mathbf{k}\rangle. \quad (1.2)$$

As said before, our  $\epsilon(\mathbf{k})$  function is simply the energy with respect to the Fermi energy:

$$\epsilon(\mathbf{k}) = E(\mathbf{k}) - E_F = \frac{\hbar^2 k^2}{2m} - \frac{\hbar^2 k_F^2}{2m}. \quad (1.3)$$

This simplifies to

$$\epsilon(\mathbf{k}) = \frac{\hbar^2}{2m}(k^2 - k_F^2) \approx \frac{\hbar^2 k_F}{m}(k - k_F). \quad (1.4)$$

A poorly drawn schematic of two electrons forming a Cooper pair is depicted on the right. This Cooper pair is precisely the kind of bound state that we have been describing. If we were to instead express our bound state as a superposition of *all*  $k$ -states (momentum states), we would have the following wavefunction:

$$|\chi\rangle = \sum_k b_k |\mathbf{k}\rangle. \quad (1.5)$$

Plugging this wavefunction into the Schrodinger equation yields

$$(\mathcal{H}_0 + U) |\chi\rangle = E |\chi\rangle, \quad (1.6)$$

where  $U$  is the attractive potential. We will give this potential the property  $U_{kk'} = -U$  for all states that are within the energy shell near the Fermi surface. This shell can be expressed as  $\Omega = \hbar v_F \Delta k$ . Under this condition,  $U$  would therefore be zero anywhere outside of the Fermi surface. By solving for our coefficients  $b_k$ , applying a

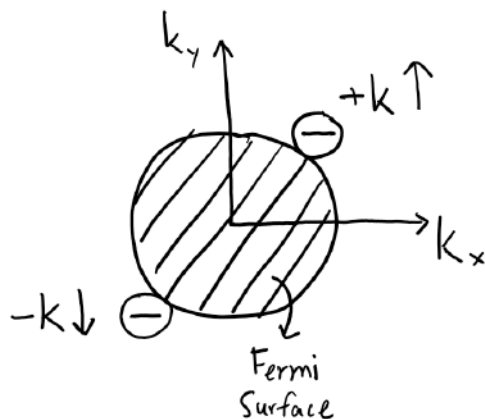


Figure 1.1: A Cooper Pair.

continuum limit, and turning our sums over momentum into an integral over the density of states at the Fermi level ( $g_F$ ), we can achieve the following self-consistency condition:

$$\frac{g_F U}{2} \ln \left( \frac{E - 2\Omega}{E} \right) = 1. \quad (1.7)$$

If we consider a scenario of weak-coupling ( $e^{e/g_F U} \gg 1$ ), we can solve for and express our pair's binding energy as

$$E \approx -2\Omega e^{-2/g_F U}. \quad (1.8)$$

The negative sign in our final result tells us that  $E < 0$ , implying that a pair of electrons will form a stable bound state for arbitrarily weak attractive interactions. This ultimately means that this pair of electrons possesses less total energy than unpaired electrons floating freely over the Fermi level!



Figure 1.2: A Cooper's Hawk.

## 2 van der Waals Crystals

For a while now in the realm of condensed matter physics, there has been a growing popularity of what is known as van der Waals materials. We can simply define **van der Waals materials** as crystals that one can mechanically decompose into atomically thin layers. The reason why this is possible can be found in the given material's lattice structure where the in-plane *covalent* bonds between atoms are much stronger than the weak *van der Waals* forces that hold the out-of-plane layers of atoms together. It turns out that all one really needs is some scotch tape to break these van der Waals forces of attraction and thus obtain atomically thin layers. I leave my definition as simple as this because it is well-known by now that certain materials carry this property. The key takeaway from all this van der Waals talk is that we can use them to accurately study the behavior of particles in solids under the 2D limit!

One of the earliest (and still ongoing) studies in such materials has been in graphene (Figure 2.1), which is most famously known to form a superconducting state (Cooper pairs wink wink) when two layers of it are stacked and twisted by some angle  $\theta$ . However, since the initial investigations of graphene, physicists have explored a great variety of other materials that also carry these weak van der Waals forces. One of the most prominently used is hexagonal boron nitride (hBN), which is most often serves as an insulating and dielectric layer that encapsulates other 2D materials. With these supporting layers, crystals like graphene can maintain high electron mobility in the case where one might want to perform some electrical experiments.



Figure 2.1: A flake of graphene.



Figure 2.2: A flake of boron nitride.

Condensed matter physicists also use hBN as a sort of protective layer that prevents air-sensitive materials from going bad. Such air-materials can be what are known as transition metal dichalcogenides (TMDs). For our sake, we will simply define these things as semiconductors (tunable band gap) that can be broken down to the 2D limit like graphene and hBN. The whole inspiring part about this van der Waals business is that one can come up with any combination of these 2D flakes and use each one's intrinsic properties to form some hybrid material or van der Waals heterostructure.

I paid a trip to the San Francisco botanical garden the other day and saw these things. It turns out that they are called ruby clusters (or *helichrysum amorginum* for fancy people). They look nice :)



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A simple black-and-white line drawing of a plant. It features several long, thin stems with clusters of small, spiky leaves or flowers at their tips. The overall shape is somewhat irregular and organic. Below the main cluster of plants, there is a single stem ending in a larger, more complex, bushy-looking structure.

birth

Figure 4.1: A drawing.



Figure 4.1: A drawing.

## 5 Back home for winter break

Cut to a wide shot of the school. Spring break is approaching and the local university students have already left. This is why the parents are home at this hour. It is still raining outside.

Cut to the inside of science class (the last class of the day). Vernon with an “o” is presenting.

Junie (Juniper) shares this class with Vernon and we can see her amongst the other students. The others are not clearly visible. Junie and Vernon are the 1001 and 1002 kids, respectively, so they often walk home together right after class. They’re not really friends.

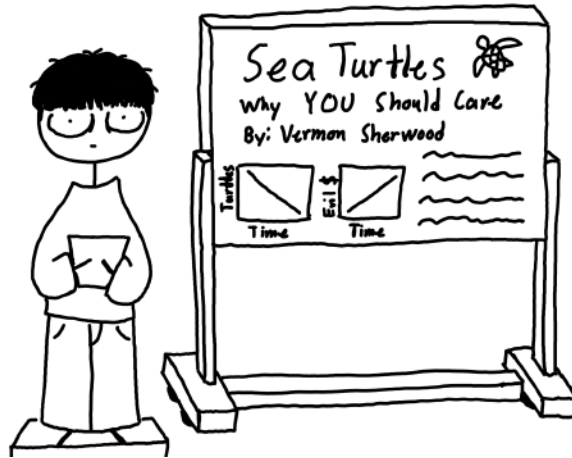


Figure 5.1: A Vernon.

## 6 Soil

I have struggled to make music lately. Hopefully, I will have more time in the summer to be creative. This small melody (Figure 6.1) is one that I wrote during winter break.



Figure 6.1: A terrible germ of an idea.



Figure 6.2: A sparrow.